Why Prediction Markets Are Broken (And How to Fix Them)

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Prediction markets are having their moment. Everyone sees their potential, but key challenges remain unsolved. This article examines recent controversies to reveal what's not working specifically regarding dispute resolution. For builders, the opportunity is clear: prediction markets are still in their early innings, and whoever solves these core issues will lead the next wave of innovation.

Introduction

Prediction markets **harness the power of financial incentives to aggregate information**. Traders bet real money on their beliefs, driving prices towards probabilities that reflect collective wisdom. When functioning properly, this process can surface more accurate forecasts than traditional methods.

This power was widely recognized in the 2024 U.S. Presidential election forecasting, where Polymarket has emerged as a more reliable indicator than traditional polling data, eventually **successfully predicting Trump's victory.**

Polymarket's credibility has grown as mainstream media, who are typically skeptical of crypto projects, began treating it as a legitimate data source. Bloomberg cited its odds in reporting, Perplexity displayed its predictions in search results, and traditional media increasingly references its data for forecasting insights.

Vitalik has also long been a fan of them, noting that "Prediction markets (and Community Notes) are becoming the two flagship social epistemic technologies of the 2020s."

Yet this **promise of algorithmic truth-seeking faces significant challenges.** A recent disputed market on Polymarket regarding the occurrence of U.S. Government Shutdown reveals critical flaws in the system, offering important lessons about the complexities of decentralized truth verification.

This article unpacks the controversy in detail and discusses the critical vulnerability in prediction market's design with regards to dispute resolution and explores potential improvements and solutions.

How does Polymarket work?

Polymarket functions like a traditional exchange, but instead of trading assets, users trade probabilities. Take the market "Will Bitcoin hit $100k in 2024?". Traders can buy or sell positions between 0% (impossible) and 100% (certain) using a familiar order book system.

If you believe Bitcoin will reach $100k in 2024 and buy $100 worth of "Yes" tokens at 47 cents, you'll receive $212 if you are correct (calculated as 100/0.47), representing the inverse of your entry price. This continuous trading allows market participants to adjust their positions as new information emerges, providing real-time insight into collective predictions.

The trading mechanics are built on what's called the Conditional Token Framework. Let's say there's $1000 in the Bitcoin prediction market:

- Alice thinks Bitcoin will hit $100k and buys $200 of "Yes" at 20 cents
- Bob is confident it won't and buys $800 of "No" at 80 cents
- The system matches these orders because they total to $1000 (100%)
- It takes $1000 USDC, creates 1000 pairs of Yes/No tokens
- Alice receives 1000 Yes tokens for her $200 (20 cents each)
- Bob receives 1000 No tokens for his $800 (80 cents each)
- When 2024 ends, winners redeem each token for $1
- If Bitcoin hits $100k, Alice's $200 becomes $1000 (5x return). Bob's token is now worth zero.

All trades happen automatically on the Polygon network, with market outcomes determined by social consensus. And in the case of a dispute of the outcome, UMA protocol, an optimistic oracle system, helps verify and determine the final resolution of the market. On a high level here is how UMA works:

- When there's a dispute about a market outcome, anyone can trigger a vote
- UMA token holders vote on the outcome
- Voting power is proportional to how many UMA tokens you hold

Vote winners get rewards, losers get penalties

Source: https://www.youtube.com/watch?v=gzbLClVAA3o

Reports from ASXN and Shoal Research offer more comprehensive explanations on the mechanics.

Controversy about US govt. shutdown case

So great, prediction markets are good at predicting event outcomes and its success in the US election should serve as a great token of credibility.

But what happens when it breaks down? A recent controversy over the U.S. Government Shutdown market reveals the current limitations in the system.

The setup seemed straightforward. Polymarket created a market on whether there would be a U.S. government shutdown between August 30 and December 31, 2024. Despite President Biden signing a funding bill (H.R. 10545 American Relief Act) **that averted a government shutdown**, along with **unanimous media consensus** across the political spectrum confirming no federal disruption occurred, the market showed a persistent 99% probability of shutdown leading into the final days of listing of the market and eventually resolved to "Yes".

What is the source of this disconnect? The crux of the issue is **related to a "rule clarification"** added by Polymarket after significant bets have been made and trading had occurred, introducing a December 20th 2024 midnight deadline that wasn't present in the original market rules. What should have been a simple binary market turned into a debate about prediction market manipulation and flaws in the design mechanics.

Chronology of events:

> Around 6pm EST on Dec 20th, odds for "yes" (meaning there will be shutdown) sat at 20%, having dropped from 70% as traders expected Senate approval of H.R.10545.

Polymarket @Polymarket · Dec 21, 2024
Odds of a government shutdown are down to just 20%.

The funding bill is going to pass.

81 67 178 36K

Then later that evening Polymarket added a banner to the market UI stating the market would resolve to "yes" if Biden didn't sign the bill by midnight. The odds for "yes" skyrocketed to 98%, as traders bet the Senate couldn't pass the bill in time to get Biden's signature.

The market's comment section exploded with heated debate. "No" holders were perplexed at the odds spike, pointing out that every news source was reporting the Senate would pass the bill shortly to avert the shutdown.

At 12:38 AM Dec 21st: Senate passed the funding bill

Morning of Dec 21st: Biden signed the bill into law, with universal media coverage confirming a shutdown was averted.

https://edition.cnn.com/politics/live-news/trump-government-shutdown-12-20-24/index.html

So why does it resolve to yes when there is no actual shutdown?

The market resolved to 'yes' despite no actual shutdown occurring. To understand how this happened, we need to examine the original market rules."

Source: https://polymarket.com/event/us-government-shutdown-before-2025?tid=1736146860643

Point 1 – fairly straightforward, observes whether a shutdown occurs between the specified period (note the latter date is Dec 31st, 2024).

Point 2 – This is the point of contention. The “yes” holder believes that the rule stipulating the president must sign the relevant bill by the applicable deadline. They believe Dec 20th midnight counts as an applicable deadline and since it was missed, this market should resolve to yes (we will get back to this in a bit).

Point 3 – covers the case about partial shutdown, not too relevant here.

Point 4 – States that the primary resolution of the market will be official info from the US government and consensus reporting.

The "Yes" Camp's Argument:

Polymarket added a banner stating midnight Dec 20th deadline

Polymarket issued ‘additional context’ on Dec 21st in favor of the Dec 20th midnight rule.

Biden didn't sign by midnight = automatic "Yes"

Rules are rules, even if no actual shutdown occurred

The "No" Camp's Argument:

First, let's examine the timing issue:

The market's original scope was clearly August to December 31st, 2024. While "Yes" voters fixate on the midnight December 20th deadline, the rules only mention "applicable deadline(s)"

> Federal appropriations operate on a daily basis, making the actual deadline 11:59pm on December 21st to avoid a shutdown.
>
> The banner saying "midnight deadline" remained visible throughout December 21st while markets were still open, which doesn’t make sense as the criteria for resolving has passed.

Second, consider the reality on the ground:

> The White House's Senior Deputy Press Secretary confirmed: "OMB has ceased shutdown preparations" due to confidence in imminent bill passage
>
> Think about what a deadline means: if missed, a shutdown would occur. No shutdown occurred, so by definition, no critical deadline was missed

Finally, A separate Polymarket on "Will House and Senate pass funding bill by midnight?" correctly resolved to "No". The point here is that a missed procedural deadline doesn't equal a shutdown, that's confusing process with outcome. That is why there are two separate pages where the spirit of the market is different.

The core tension here isn't just about interpretation. It's about whether prediction markets should prioritize technical rule-reading over real-world outcomes they're supposedly designed to predict. When a market resolves "Yes" to a shutdown that objectively never happened, something has gone wrong with the truth-seeking mechanism.

Disputes like this are not uncommon

One may react to this as just one isolated incident as a result of poorly written rules. But controversies like this are not uncommon. A watchdog site, Polymarketfraud (excuse the provocative name), documents numerous cases where market resolutions have contradicted reality.

The **Venezuela Presidential election winner** market is particularly interesting. The current president in Venezuela is Nicolas Maduro yet the market resolved that opposition candidate Edmundo Gonzalez had won in the most recent election.

Frank Muci has explored this in more depth in this article. Here is a quick summary.

> The market rules stated: "The primary source of resolution is official information from Venezuela, however a consensus of credible reporting will also suffice"
>
> Official results showed Maduro winning:
>
> First announcement: 51.20000% vs 42.20000%
>
> Second announcement: 51.9500% vs 43.1800% (The suspicious

precision with multiple zeros suggested fabrication)

However, opposition vote tallies showed them leading by 20%+ based on polling station data.

UMA token holders (who have final say on disputed resolutions) were lobbied heavily to ignore the official Venezuelan sources and instead rely on credible media consensus about election fraud.

The UMA holders ultimately voted to override Polymarket's stated primary resolution source, effectively ruling that Gonzalez won - even though Maduro remained in power.

This inconsistency in resolution approach reveals a problem. In the U.S. shutdown case, **UMA voters adhered to a technical rule** (a late-added banner about midnight deadline), ignoring **consensus reporting** that no shutdown occurred. Yet in the Venezuelan election, they did the exact opposite, overriding the primary resolution source in favor of consensus reporting about election fraud.

Source: https://polymarketfraud.com/

Another case involved the **Israel-Hezbollah Ceasefire market**. The market resolved to "yes" even though there were credible reports of ongoing military actions. This YouTube video offers a detailed walkthrough of what happened: Gaming Prediction Markets: A $40M Lesson

In addition, **Lou Kerner** presents an intriguing theory about the US election market in his article. Though he labels it a "conspiracy theory," his analysis suggests Polymarket's presidential election market might be structurally biased toward Trump.

The scenario he envisions is basically: if Trump loses, he may refuse to concede, claiming voter fraud and disputing the election results just as he did in 2020. Therefore even if Kamala Harris effectively wins the election, the market might not resolve in her favor.

This creates what Kerner calls a "heads I win, tails I don't lose" scenario for Trump positions. If Trump wins, bettors win. If he loses but disputes the results, the market resolution could be delayed or altered through UMA token holder votes.

Issues this surfaced

First is the problem of rule manipulation. Oracles become mere decoration when platforms can add clarifications at will. In the shutdown case, market odds surged toward "yes" immediately after a new banner was posted, changing the effective deadline from December 31 to December 20, 2024.

This leads to other questions about resolution criteria. When rules conflict, which takes precedence? While the primary resolution criteria specified news sources and credible reporting with a December 31 deadline, the market ultimately resolved based on a later-added December 20 midnight clarification. This inconsistency in rule hierarchy undermines the market's credibility.

A structural challenge lies in the relationship between UMA holders and Polymarket's resolution system. Since UMA token holders can participate in both trading and voting, there is a huge incentive alignment between large traders and oracle voters.

While Polymarket and UMA were meant to be independent systems providing mutual checks, UMA is de facto Polymarket's only oracle provider. This reminds me of that scene in The Big Short where the rating agency employee admits they have to give AAA ratings or the banks will simply go to their competitor. When your success depends on keeping powerful players happy, independence becomes impossible.

Dispute resolution - prediction market’s achilles heel

A prediction market is only as good as its ability to determine truth. You can have perfect UI, sophisticated trading systems and deep liquidity, but none of that matters if you can't reliably determine who won a bet. Polymarket uses UMA's oracle system for this, but there's a potential vulnerability in how it works.

Let’s recap on the basic UMA mechanics:

When there's a dispute about a market outcome, anyone can trigger a vote

UMA token holders vote on the outcome

Voting power is proportional to how many UMA tokens you hold

Vote winners get rewards, losers get penalties

Luca Prosperi from Dirt Roads articulates the oracle problem of polymarket in his blog. In it he puts forth a concept called the **Corruption Value Multiple (CVM):**

Polymarket has about $300M in open bets

UMA's total market value is around $220M

You'd need about $110M to control half of UMA's tokens

This means for every $1 spent controlling UMA, you could influence $1.36 n bets

But the real situation might be riskier because:

Only about 20% of UMA tokens typically vote, not 100%

Market rules are often ambiguous

There's room for social manipulation

The actual cost to influence outcomes might be much lower than $110M

It means traders might rationally bid prices far above true probabilities if they believe they can influence the oracle's decision.

These are difficult design problems, and I don't pretend to have a silver bullet. However, getting dispute resolution right is perhaps the most critical challenge facing prediction markets. When outcomes are arbitrated inconsistently, users lose faith in the system. Over time, this inconsistency pushes the markets away from their intended purpose.

Potential fixes and improvement

What can be done to improve the situation? Some ideas:

Once a market goes live, its rules should be set in stone. A market's terms shouldn’t be unilaterally modified after creation, prediction markets shouldn't allow post-listing "clarifications." Instead, the original market rules should serve as the complete and final reference document. When disputes arise, oracles should interpret these base rules without influence from platform-added clarifications or banners.

Establish rule hierarchy and on-chain proofs. Market rules need clear precedence order (i.e. which rules override others when conflicts arise). Primary resolution criteria (like consensus reporting) should explicitly outrank

secondary mechanisms. This hierarchy must be recorded on-chain at market creation, creating an immutable proof of not just the rules themselves, but their relative authority.

Diversifying oracle systems. Markets could require multiple independent verification systems, each with different mechanisms and stakeholders. This creates redundancy and makes manipulation more costly and complex. Multiple oracles can cross-verify outcomes, reducing reliance on one system.

Incorporate reputation-based verification. Beyond token voting, markets could implement a council system where respected industry figures stake their professional reputation to decide outcomes. This adds social accountability and expertise to the verification process.

Intersubjective forking. Building on Eigenlayer's innovation, this mechanism handles cases where human consensus recognizes clear wrongs. When major disputes arise, the community could split the token (whether its oracle or protocol token) used for resolution into two versions, one supporting each interpretation of the outcome. Market forces would then determine which fork retains value. Those who back the wrong interpretation would see their tokens lose value, creating natural economic penalties for supporting manipulated outcomes.

AI agent as an impartial arbiter. What if, instead of relying on token holders who have financial incentives to manipulate, we trained a specialized AI agent with the single purpose of deciding outcome? Unlike human voters who might vote based on their positions, an AI agent could be specifically designed to analyze evidence impartially and arrive at more accurate market resolutions.

Conclusion

To be clear, this post isn't meant as criticism of Polymarket specifically. But as the largest (and let’s be honest) the only relevant player in crypto prediction markets right now, it serves as the best case study for understanding sector wide challenges.

Why does all this matter? If you view prediction markets merely as gambling platforms where traders bet on outcomes, their flaws would be of limited consequence. Sure some people lost some money but in the end it’s just another venue for speculation.

But prediction markets are increasingly positioned as truth engines, objective barometers of reality that can cut through noise and bias to reveal genuine probabilities of future events.

This is why the government shutdown case matters. When a market confidently predicts and validates a shutdown that objectively never occurred, it reveals how these truth engines can manufacture their own

alternate reality. The danger isn't just that some traders lost money. The danger is that we're building supposedly objective verification systems that can be captured by those with the capital and incentive to shape public perception.

As prediction markets grow more influential, their structural weaknesses become everyone's problem. Either we solve their fundamental vulnerabilities or we risk creating powerful tools for distorting rather than discovering truth.

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